Amit Yadav

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Summary

AI/ML engineer and **AI training expert** with hands-on experience on **3 Mercor AI training projects** – Gujarati document annotation and transcription, Gujarati LLM response evaluation and adversarial prompting (red-teaming), and designing, writing and recording narrated professional computer tasks for AI training data. Research experience in **model evaluation**, benchmarking, and comparative analysis across NLP and computer vision tasks. Practitioner of AI coding agents (Claude Code, Cursor, GitHub Copilot). Experienced in assessing model outputs for correctness, failure modes, and edge cases, following detailed annotation guidelines precisely, and delivering work that clears automated QC and peer review. Fluent in English and Gujarati.

Education

M.Tech (Data Science and Engineering) – NIT Jamshedpur 2025 – Present CGPA: 8.19/10
B.E (Computer Engineering) – GEC Dahod (G.T.U) 2021 – 2025 CGPA: 8.05/10

AI Training & Data Annotation Experience

Mercor – AI Training Expert (Contract, Remote) Sep 2026 – Nov 2026
3 projects across task writing, LLM evaluation & red-teaming, document annotation, and Gujarati transcription

Odin – Writer Sep 2026 – Nov 2026

- Designed realistic, multi-step **professional computer tasks** and wrote full task specifications (domain, goal, starting materials, steps, deliverable) used to build AI training data.
- Wrote first-person **narration scripts** and recorded narrated screen workflows explaining every action and the reasoning behind it, closing with a structured review of approach, challenges, and quality.
- Prepared Python scripts and input data in advance, verified every reported metric by running the code, and packaged deliverables to strict naming and format rules; met **automated QC limits** and gained reviewer approval.

PDF Annotator – Gujarati Sep 2026 – Oct 2026

- Annotated the **layout of Gujarati document pages**: outlined regions (titles, headings, tables, figures, answer fields), assigned component types and reading order, and linked captions and notes to their parent elements.
- **Transcribed printed and handwritten Gujarati** text in native script, flagged illegible regions, recorded page metadata (document type, language, table/formula/handwriting flags), and screened pages for legibility and personal information; work verified through peer review.

Generalist – Gujarati – Annotator Sep 2026 – Oct 2026

- Crafted **adversarial Gujarati prompts** designed to push LLMs into critical failures, probing reasoning, factual accuracy, instruction following, and language handling.
- Performed **LLM response evaluation** against project guidelines, identifying and documenting model failures and judging outputs for correctness, fluency, natural register, and cultural context.

Work Experience

Research Intern – Machine Vision and Intelligence Lab, NIT Jamshedpur Oct 2025 – Present
Under the supervision of Dr. K. K. Singh

- **Evaluated and compared frontier NLP models** (mBERT, MuRIL, custom Hindi tokenizer) across 3 structured evaluation tracks – text classification, intrinsic tokenization metrics, and NER – applying systematic benchmarking methodology analogous to AI model assessment workflows.
- Identified **failure modes and performance gaps** in multilingual pretrained models; quantified OOV rate deficiencies (39.27% for mBERT vs. 8.98% for monolingual model) and documented architectural weaknesses across Devanagari language tasks.
- Assessed **model-generated outputs** across 5 Hindi NLP datasets for quality and correctness; designed evaluation rubrics covering precision, recall, F1, and zero-shot transfer – skills directly applicable to evaluating AI model outputs and AI-generated code.
- Demonstrated that a **6.6× smaller vocabulary** monolingual tokenizer outperforms MuRIL across all 3 evaluation tracks; distilled findings into actionable engineering recommendations for production deployment.
- Worked on **Noble Nudity Detection** using computer vision; scraped, annotated, and curated image datasets from multiple sources for supervised model training.

Projects

FaceRoom AI: Distributed Face-Based Photo Retrieval System GitHub

- Built a full-stack event photo management platform; used **AI coding agents (Claude Code, Cursor)** throughout the SDLC – reviewed agent-generated code for correctness, maintainability, and architectural soundness before integration.
- Designed **RESTful backend** with FastAPI, JWT authentication, room management, image metadata handling, and PostgreSQL integration; tracked development workflow using **JIRA** under Agile/SDLC practices.
- Implemented **async pipelines** with Redis and Celery workers for face detection and ArcFace embedding-based clustering; built **Qdrant vector search** – selfie upload returns matching event photos via similarity search.

- Containerized full application using **Docker** orchestrating API server, Celery workers, PostgreSQL, object storage, and vector search; evaluated performance, scalability, and edge cases throughout.

Renal Malignancy Detection using Classical ML & Deep Learning GitHub

- **Compared multiple model architectures** (SVM, LR, Random Forest, ResNet18 Transfer Learning) with rigorous evaluation; identified that classical ML with GLCM features outperformed deep learning baseline (**99% accuracy, 96% tumor recall** vs. 93% DL) – demonstrating strong model comparison and failure-mode analysis skills.
- Deployed interactive **Streamlit** web app for real-time benign/malignant prediction with image upload, preprocessing, and feature extraction pipeline.

Osteoporosis Screening using Fibonacci Feature Engineering GitHub

- Built ML pipeline for knee X-ray analysis using Fibonacci-based segmentation; extracted 90 handcrafted features; evaluated using precision, recall, F1-score, and ROC-AUC.

Amazon ML Hackathon: Multi-Modal Price Prediction GitHub

- Predicted product prices via **CLIP text embeddings** + **EfficientNet image features**; PCA + LightGBM achieved **SMAPE: 59.36** – AIR 1963 among 83,000+ teams.

Skills

AI Training & Annotation: LLM Response Evaluation, Adversarial Prompting & Red-Teaming, Data Annotation & Labeling, Document Layout Annotation, Transcription (printed & handwritten Gujarati), AI Task Design & Specification Writing, Narrated Workflow Recording, Guideline Adherence, QC & Self-Review
Spoken Languages: English, Gujarati
AI Coding Agents: Claude Code, Cursor, GitHub Copilot, Windsurf, Gemini CLI, Codex
Model Evaluation: Comparative Benchmarking, Failure Mode Analysis, Edge Case Identification, Output Quality Assessment
Programming: Python, C/C++, SQL, JavaScript, Swift (learning)
ML/DL: Regression, Classification, Feature Engineering, Model Evaluation, Transfer Learning, BiLSTM-CRF
NLP: Tokenization, WordPiece, BPE, SentencePiece, TF-IDF, Bag-of-Words, NER, Sequence Labeling
CV: Image Processing, GLCM Feature Extraction, Face Detection, ArcFace Embeddings, EfficientNet, ResNet
Backend & DevOps: FastAPI, Flask, PostgreSQL, Redis, Celery, Qdrant, Docker, REST APIs, JWT Auth
Tools: Scikit-learn, HuggingFace Tokenizers, Pandas, NumPy, Matplotlib, XGBoost, Streamlit, Seaborn
Practices: Agile, SDLC, JIRA, Git, GitHub, LaTeX, Web Scraping, Jupyter, Anaconda

Achievements

- Completed **3 AI training projects on Mercor** (Sep – Nov 2026) across task writing, LLM evaluation, annotation, and Gujarati transcription
- Qualified **GATE 2025** – 97.15 percentile
- **Amazon ML Challenge 2025** – AIR 1963 among 83,000+ registered teams
- Research Intern at Machine Vision and Intelligence Lab, NIT Jamshedpur (under Dr. K. K. Singh)